AI-Powered Automatic Data Analysis

(Smart analytics for smarter decisions)

National Institute of Electronics and Information Technology

SRINAGAR CAMPUS

SIDCO Electronics Complex Rangreth Old Airport,

Srinagar, Kashmir

Affiliated to

University of Kashmir

Department of Computer Science

AI-Powered Automatic Data Analysis

(Smart analytics for smarter decisions)

PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

MCA (Masters of Computer Applications)

in

Department of Computer Science

University of Kashmir

National Institute of Electronics and Information Technology

BONAFIDE CERTIFICATE

Certified that this project report titled

"AI-Powered Automatic Data Analysis"

is the bonafide work of Mr. Waqar Maqbool and Mrs. Inshpreet Kaur Mehta, who carried out the project work under my supervision and guidance in the **Department of Computer Science** during the academic year **2025**.

This work has not been submitted previously for the award of any degree, diploma, or certificate at any other university or institute.

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DECLARATION

We hereby declare that the project report titled **"AI-Powered Automatic Data Analysis"** is an original work carried out by us as a part of our academic curriculum in the **Department of Computer Science**.

This work has been conducted under the guidance of **Dr. Syed Nisar Bukhari (Scientist-D)**, and we affirm that it has not been submitted previously to any other university or institution for the award of any degree, diploma, or certificate.

We also confirm that the content presented in this report is the result of our own efforts and research. Proper acknowledgments have been given wherever the work of others has been cited.

We understand that any form of plagiarism or misrepresentation may lead to disciplinary action as per institutional policies.

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ABSTRACT

In the big data age, it is still very much a lengthy and laborious process of transforming raw datasets into meaningful insights, particularly for non-technical users. This project, AI-Powered Automatic Data Analysis undertakes to bridge that gap with an easy, web-based system whereby users can upload their datasets and then conduct full data analysis through Natural Language Processing (NLP).

The system is built using Python and Flask, with Pandas to do the main data processing work. It allows data upload in several formats- CSV, Excel, JSON, and it will immediately do cleaning, summarization, anomaly detection, and visualization. A big part of the platform is an AI-powered chatbot included through OpenRouter API which helps users talk to their data using simple English questions. This makes big analysis work like filtering, adding up and finding patterns easier without having to write code.

Users can explore the data through an interactive dashboard, generate charts, and apply filters in real-time. There is also a feedback and review module to collect user inputs for helping to improve the tool iteratively.

The architecture has been designed modular and extensible to enable the easy integration of future Machine Learning tools and services. It aims at democratizing data analytics, making it accessible to a wider audience, and more importantly, non-technical backgrounds.

This report documents the system's design, implementation, key features, and potential future enhancements. It presents an in-depth view of how Artificial Intelligence can simplify as well as improve the analysis of data.

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